

Qatar-1b

R-2174

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_121015 · created 2026-08-13T00:52:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456215.64262 +/- 0.00011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.116 +/- 0.026
Transit depth (Rp/Rs)^2	1.35 +/- 0.61 [%]
Orbital Inclination (inc)	84.98 +/- 1.93
Airmass coefficient 1 (a1)	1.06 +/- 0.016
Airmass coefficient 2 (a2)	-0.045 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	None
Optimal Aperture	2.79
Optimal Annulus	5.44
Transit Duration (day)	0.068 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.116 ± 0.026 vs 0.1463 (deviation 0.8σ)
Duration fit vs published	0.068 d vs 0.0692 d (deviation 0.07σ)
Mid-time O–C	-0.3 min
Depth SNR	3.1
Residual scatter	1.51 %

Light curve

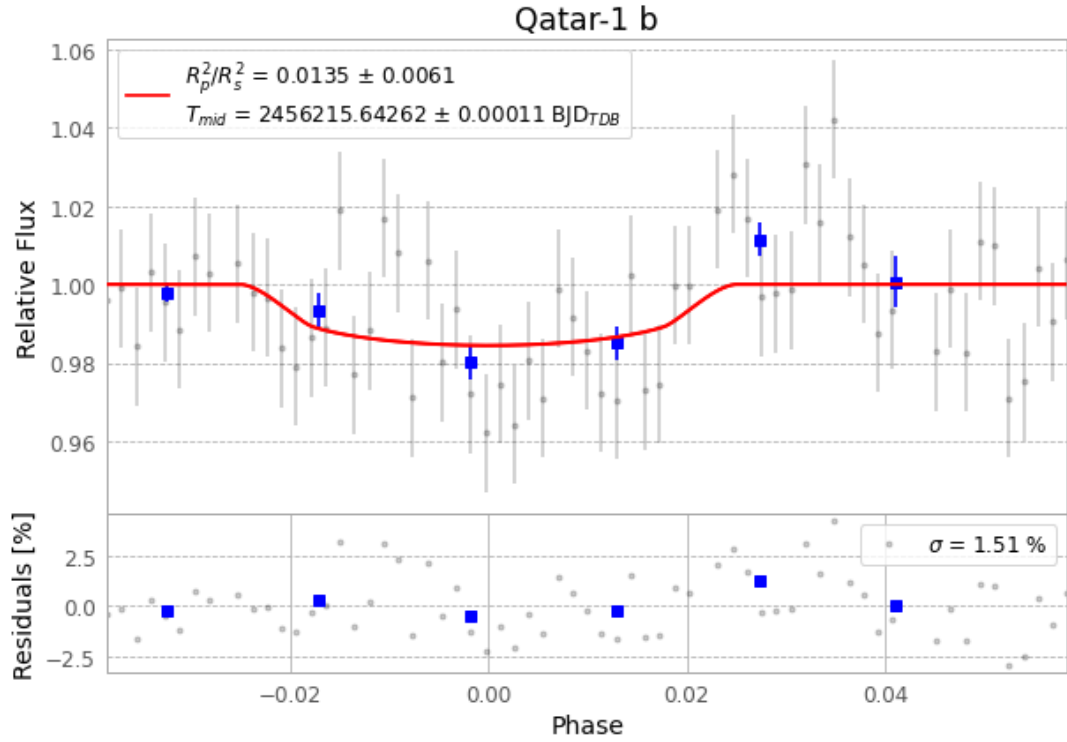


Figure 1 — FinalLightCurve_Qatar-1 b_2012-10-14.png (SHA-256 239661f4df81b9a0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [429, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5bc73b09fc954f60...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), Qatar-1121015020312.FITS → Qatar-1121015052112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-121015.log (8ac535203412...), FinalLightCurve_Qatar-1 b_2012-10-14.png (239661f4df81...), FinalLightCurve_Qatar-1 b_2012-10-14.csv (190877381a42...)

Compute resources

Wall time 143.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:52Z.